

README

Migrants at Sea: Unintended Consequences of Search and Rescue Operations

Overview

The code in this replication package constructs the material files using STATA.

The main folder, namely “Replication”, contains 5 subfolders and 2 files. The 5 subfolders are:

- **data** which contains the list of the dataset in .dta format (STATA) and a sub-subfolder with the data necessary to create the map (such as Figure C4);
- **do** which contains: (i) a dofile for installing the packages (“dopackages.do”) and (ii) a master do-file that creates all figures and tables in .png and .tex formats, respectively (“dofile.do”);
- **graphs** which contains the figures of the manuscript. The users need to run the dofile.do;
- **List of tables and figures** which contains a .tex file (“list of tables and figures”) that uploads all materials (figures and tables) included in the manuscripts.

The two documents are:

- the README file (**README.pdf**).
- the **manuscript**.

The Replication Code section provides instructions for running the replication code. The Data Availability Statement section lists all the data sources, describes how researchers and users can access the data and provides download sources.

The replicator should expect the code to run for less than 60 minutes.

Data Availability and Provenance Statements

The manuscript uses mostly confidential data sources although some publicly available datasets are also included. The dataset by *Polizia di Stato* (the Italian State Police) and *Frontex* (EU border agency) are confidential and the users need to contact the national authorities to receive their own version. These data cannot be made publicly available. The authors will assist with any reasonable replication attempts by the users. Nevertheless, it can take some time to negotiate data use agreements and gain access to the data. Below we describe the details. When the dataset is publicly available we provide the link to facilitate the download.

Details on Data Sources

For each figure and table presented in the manuscript, we provide a description of the data source, how we cite them in the paper, and the corresponding steps to obtain them, if not publicly available.

- **Figure 1** exploits the yearly Italian data on total crossings that come from the *Polizia di Stato*, the Italian State Police.

[*Citation*: Polizia di Stato. 2009–2020. “Data On The Total Number of Crossings to Italy”, Ministry of Interior.]

In order to receive similar or additional data, the users can email `dipps.servizioimmigrazione.2div@interno.it`, which is the Central Directorate for Immigration and Border Police that deals with the fight against illegal immigration, with the management of activities relating to the granting of residence permits to foreign citizens and with the supervision and control of land, naval and airport borders of Italy. It can take 15 working days to negotiate data and additional 15 days to gain access to the data.

The data on deaths at sea are from:

UNITED for Intercultural Action available at <https://unitedagainstrefugeedeaths.eu/>.

[*Citation*: List of Deaths. 2009–2020. “Data On The Total Number of Deaths and Missings to Italy”, <https://www.unitedagainstrefugeedeaths.eu/>.]

The Migrants File data available at <https://www.themigrantsfiles.com>.

[*Citation*: The Migrants File. 2009–2016. “Data On The Total Number of Deaths and Missings to Italy”, <https://www.themigrantsfiles.com>.]

- **Figure 2** exploits the information on crossing attempts, which are the sum of crossings and deaths in transit, are disclosed by the European Border and Coast Guard Agency known as *Frontex* (for the period from 1 January 2013 to October 2017) and by *Polizia di Stato* (for the period November 2017 to December 2020).

In order to get the *Frontex* dataset, the users can email the European Border and Coast Guard Agency at <https://www.asktheeu.org/> which is the website that sends the request to an EU body. All answers are public so that the information is shared with anyone who is interested. The users should get a reply within 15 working days (3 weeks). Then additional 15 days to get the data.

[*Citation*: Frontex. 2013–2017. “Data On The Type of Boats”, European Border and Coast Guard Agency, European Union.]

In order to receive the data by *Polizia di Stato*, the users can email *dipps.servizioimmigrazione.2div@interno.it*, which is the Central Directorate for Immigration and Border Police that deals with the fight against illegal immigration, with the management of activities relating to the granting of residence permits to foreign citizens and with the supervision and control of land, naval and airport borders of Italy. It can take 15 working days to negotiate data and additional 15 days to gain access to the data.

[*Citation*: Polizia di Stato. 2016–2020. “Data On The Total Number of Crossings to Italy by Type of Boats and Routes”, Ministry of Interior.]

- **Figure 3** exploits the information on crossing attempts, which are the sum of crossings and deaths in transit, by different routes (Libya and Tunisia) and types of boats (Inflatable and Not Inflatable boats).

In order to receive the data by *Polizia di Stato*, the users can email *dipps.servizioimmigrazione.2div@interno.it*, which is the Central Directorate for Immigration and Border Police that deals with the fight against illegal immigration, with the management of activities relating to the granting of residence permits to foreign citizens and with the supervision and control of land, naval and airport borders of Italy. It can take 15 working days to negotiate data and additional 15 days to gain access to the data.

[*Citation*: Polizia di Stato. 2017–2020. “Data On The Total Number of Crossings to Italy by Type of Boats and Routes”, Ministry of Interior.]

- **Figure 4** and **C1** exploits the info the *Frontex* dataset, the users can email the European Border and Coast Guard Agency at <https://www.asktheeu.org/> which is the website that sends the request to an EU body.

In order to get the *Frontex* dataset, the users can email the European Border and Coast Guard Agency at <https://www.asktheeu.org/> which is the website that sends the request to an EU body. All answers are public so that the information is shared with anyone who is interested. The users should get a reply within 15 working days (3 weeks). Then additional 15 days to get the data.

[*Citation*: Frontex. 2013–2017. “Data On The Type of Boats”, European Border and Coast Guard Agency, European Union.]

- **Table 2** show the summary from:
 - (i) the dataset from *Frontex*. In order to get the *Frontex* dataset, the users can email the European Border and Coast Guard Agency at <https://www.asktheeu.org/> which is the website that sends the request to an EU body. All answers are public so that the information is shared with anyone who is interested. The users should get a reply within 15 working days (3 weeks). Then additional 15 days to get the data.

[*Citation:* Frontex. 2013–2017. “Data On The Type of Boats”, European Border and Coast Guard Agency, European Union.]

(ii) dataset from *Polizia di Stato*, the Italian State Police (Sample 2016–2020) exploits the information on crossing attempts, which are the sum of crossings and deaths in transit, by different routes (Libya and Tunisia) and types of boats (Inflatable and Not Inflatable boats).

In order to receive the data by *Polizia di Stato*, the users can email dipps.servizioimmigrazione.2div@interno.it, which is the Central Directorate for Immigration and Border Police that deals with the fight against illegal immigration, with the management of activities relating to the granting of residence permits to foreign citizens and with the supervision and control of land, naval and airport borders of Italy. It can take 15 working days to negotiate data and additional 15 days to gain access to the data.

[*Citation:* Polizia di Stato. 2016–2020. “Data On The Total Number of Crossings to Italy by Type of Boats and Routes”, Ministry of Interior.]

(iii) Weather conditions (“Wave Height”) come from ERA5 which is the fifth generation of European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis for the global climate and weather for the past 4 to 7 decades. Currently data is available from 1950. The data can be downloaded from <https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels?tab=form>

- **Table 3, 5, C3-C9; Figure 8, C4, C8-C10, and D1** exploit the dataset from:

(i) *Frontex*. In order to get the *Frontex* dataset, the users can email the European Border and Coast Guard Agency at <https://www.asktheeu.org/> which is the website that sends the request to an EU body. All answers are public so that the information is shared with anyone who is interested. The users should get a reply within 15 working days (3 weeks). Then additional 15 days to get the data.

[*Citation:* Frontex. 2013–2017. “Data On The Type of Boats”, European Border and Coast Guard Agency, European Union.]

(ii) Weather conditions (“Wave Height”) come from ERA5 which is the fifth generation of European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis for the global climate and weather for the past 4 to 7 decades. Currently data is available from 1950. The data can be downloaded from <https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels?tab=form>

- **Table 4, C1 and Figure 9, C2, C11** exploit dataset from:

(i) *Polizia di Stato*, the Italian State Police (Sample 2016–2020) exploits the information on crossing attempts, which are the sum of crossings and

deaths in transit, by different routes (Libya and Tunisia) and types of boats (Inflatable and Not Inflatable boats).

In order to receive the data by *Polizia di Stato*, the users can email dipps.servizioimmigrazione.2div@interno.it, which is the Central Directorate for Immigration and Border Police that deals with the fight against illegal immigration, with the management of activities relating to the granting of residence permits to foreign citizens and with the supervision and control of land, naval and airport borders of Italy. It can take 15 working days to negotiate data and additional 15 days to gain access to the data.

[*Citation*: Polizia di Stato. 2016–2020. “Data On The Total Number of Crossings to Italy by Type of Boats and Routes”, Ministry of Interior.]

(ii) Weather conditions (“Wave Height”) come from ERA5 which is the fifth generation of European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis for the global climate and weather for the past 4 to 7 decades. Currently data is available from 1950. The data can be downloaded from <https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels?tab=form>

- **Figure A1** uses data from the United Nations High Commissioner for Refugees. Data are available at <https://reliefweb.int/sites/reliefweb.int/files/resources/61535.pdf>.

[*Citation*: UNHCR, 2017. Libya: Activities at Disembarkation. Technical Report. United Nations High Commissioner for Refugees, URL: <https://reliefweb.int/sites/reliefweb.int/files/resources/61535.pdf>.]

- **Figure C3** uses the data come from UN Comtrade over the period 2005–2020. See the link <https://comtrade.un.org/data> to download similar data.
- **Figure C5** uses weather conditions (“Wave Height”) come from ERA5 which is the fifth generation of European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis for the global climate and weather for the past 4 to 7 decades. Currently data is available from 1950. The data can be downloaded from <https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels?tab=form>

Computational requirements

The code that generates the tables and the figures in the manuscript is done using STATA software. The code was run using STATA 17 on a MacBook with macOS version 12. Before using the master “dofile.do” that produces the materials in the paper, the users need to install different packages. Indeed, the users should first run the “dopackages.do” containing all programs needed.

Software Requirements

The code needs the installation of different STATA programs. The “dopackages.do” file installed a series of packages such as (**ftools**, **reghdfe**, and **ppmlhdfc**). The **ppmlhdfc** implements Poisson pseudo-maximum likelihood regressions (PPML) with multi-way fixed effects while **reghdfe** fits a linear or instrumental-variable regression absorbing a potential large number of categorical factors and interactions. Moreover, the researchers need to install three different packages (**spmap**, **shp2dta** and **mif2dta**) for the map (Figure C4). All of these - and many other - are installed by running the “dopackages.do” code in the folder “do”. It is commented.

Instructions to Replicators

The replication steps are:

1. Unzip and download the replication archive.
2. Run the “dopackages.do” to install the packages needed for the analysis.
3. Define the root directory in `dofile.do` to the file path pointing to the location of the downloaded and unzipped replication archive. In the `dofile.do` the users need to modify the `cd : "/Replication"` in row 63.
4. Run `dofile.do` in STATA. After installing the necessary programs and setting the global directories, it will reproduce the material in the **List of tables and figures.tex**. The `dofile.do` runs tables and figures following the order presented in the manuscript:

- Figure 1: Crossings and Deaths Along the Central Route, 2009-2020.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 2 minutes
- Table 1: EU Operations - no data involved. Authors elaboration.
- Figure 2: Types of Vessels Used, 2013-2020.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 2 minutes
- Figure 3: Attempted Crossings by Ports of Departure, 2016-2020.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 2 minutes
- Figure 4: Predicted Points of Departure, 2011-2017.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 2 minutes
- Table 2: Summary Statistics.

- Approximate time needed to reproduce the analyses on a standard desktop machine: less than 2 minutes
- Figure 5: Migrant’s Crossing Decisions - no data involved. Authors elaboration.
- Figure 6: Ex Ante Crossing Risk - no data involved. Authors elaboration.
- Figure 7: Incidence of Search and Rescue Operations on Migrants - no data involved. Authors elaboration.
- Table 3: Elasticities of Crossing Attempts to Crossing Conditions.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 10 minutes
- Table 4: Elasticities of Crossing Attempts to Crossing Conditions by Country of Departure.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 10 minutes
- Figure 8: Simulated Probability of Success on Unsafe Boat vs. Safe Boat by Relative Price.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 2 minutes
- Table 5: Fraction of Migrants by Boat Types.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 15 minutes
- Figure 9: Fraction of Inflatable Boats around the Introduction of the “Minniti Code”.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 2 minutes

Appendix A

- Figure C.1: Nationalities of Migrants on the Central Route by Year.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 2 minutes

Appendix C

- Figure C.1: Nationalities of Migrants on the Central Route by Year.

- Approximate time needed to reproduce the analyses on a standard desktop machine: less than 2 minutes
- Figure C.2: Monthly Crossing Attempts.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 2 minutes
- Figure C.3: Net-Imports of Rubber Boats, Wooden Ferries and Life Jackets.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 2 minutes
- Figure C.4: Map of Angles.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 5 minutes
- Table C.1: Irregular Migration During Search and Rescue Operations.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 15 minutes
- Table C.2: Wave and Swell Explanations - no data involved. Authors elaboration.
- Table C.3: Crossing Attempts: Robustness using OLS.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 10 minutes
- Table C.4: Crossing Attempts: Robustness on Cluster Standard Errors.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 15 minutes
- Table C.5: Crossing Attempts: Robustness on Wave Height.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 15 minutes
- Table C.6: Crossing Attempts on Wave Height Squared.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 15 minutes
- Table C.7: Crossing Attempts with Different Locations.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 15 minutes
- Table C.8: Crossing Attempts on Bad Weather Dummies.

- Approximate time needed to reproduce the analyses on a standard desktop machine: less than 15 minutes
- Table C.9: Fraction of Migrants by Boat Types: Robustness.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 10 minutes
- Figure C.5: Density of Wave Height in Tripoli by Season.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 2 minutes
- Figure C.6: A Typical Inflatable Boat - no data involved. Picture taken from www.alibaba.com.
- Figure C.7: Probability of Encountering Large Waves - no data involved. Authors elaboration.
- Figure C.8: Randomization Inference Using Former Wave Conditions.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 10 minutes
- Figure C.9: Randomization Inference Using Future Waves.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 10 minutes
- Figure C.10: Crossing Attempts to Crossing Conditions along Fraction of Unsafe Boats.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 10 minutes
- Figure C.11: Event Study around the Minniti Code.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 15 minutes

Appendix D

- Table D.1: NGO Vessels and Operational Period - no data involved. Authors elaboration.
 - Figure D.1: Rescue Activity by Organization 2014-2017.
 - Approximate time needed to reproduce the analyses on a standard desktop machine: less than 2 minutes
5. Once the dofile.do ends, the figures and the tables are save in the **graph** and **tables** subfolders, respectively. Only two tables need to be modified by hand. In detail, the users need to copy-paste the output from STATA in the corresponding tex table. This is the case for Table 2 with the statistics and

Table C4 with the different cluster standard errors in brackets (see Table C4) that need to be copy-paste by hand from tabC4a.tex to tabC4d.tex (folder “tables”).

6. Then the users can run the tex file “The list of tables and figures” that will show the figures and the tables of the manuscript.

References

- ECMWF, 2009–2020. Data On Weather Conditions - Wave Height . Technical Report. European Centre for Medium-Range Weather Forecasts. URL: <https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels?tab=form>.
- Frontex, 2013–2017. Data On The Type of Boats. Technical Report. European Border and Coast Guard Agency, European Union.
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- Polizia di Stato, 2009–2020. Data On The Total Number of Crossings to Italy. Technical Report. Polizia di Stato, the Italian State Police, Ministry of Interior.
- Polizia di Stato, 2016–2020. Data On The Total Number of Crossings to Italy by Type of Boats and Routes. Technical Report. Polizia di Stato, the Italian State Police, Ministry of Interior.
- The Migrants File, 2009–2016. Data On The Total Number of Deaths and Missings to Italy. Technical Report. IOM. URL: <https://www.themigrantsfiles.com>.
- UNHCR, 2017. Libya: Activities at Disembarkation. Technical Report. United Nations High Commissioner for Refugees. URL: <https://reliefweb.int/sites/reliefweb.int/files/resources/61535.pdf>.